WORKED SOLUTION

STEP I 1999, Question 5

Pure • Small Angle Approximations

STEP I 1999, Question 5

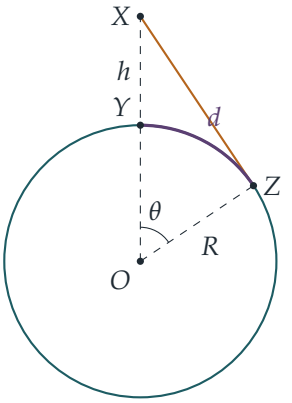
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q5

The key idea

"Just seen on the horizon" means the line of sight from Z to X grazes the Earth, so it is a tangent at Z and is perpendicular to the radius OZ. That gives a right-angled triangle with the centre O, and the arc YZ is R times the angle at O. The two approximations then come from looking at that angle when it is small, and when it is close to a right angle.

Solution



Part (i)

Let O be the centre of the Earth and let $\theta = \angle YOZ$. Since X is directly above Y, the points O, Y, X are collinear with $OX = R + h$. Since X is just visible from Z, the line ZX touches the Earth at Z, so it is perpendicular to the radius OZ. In the right-angled triangle OZX,

$$\cos\theta = \frac{OZ}{OX} = \frac{R}{R+h}.$$

The arc YZ of a circle of radius R subtending angle θ at the centre has length $R\theta$, so

$$d = R\arccos\left(\frac{R}{R+h}\right).$$

Here $0 < \theta < \frac{\pi}{2}$, since $0 < R/(R+h) < 1$.

Part (ii)

When h is small compared with R, $\cos\theta = R/(R+h)$ is close to 1, so θ is small and we may use $\cos\theta \approx 1 - \frac{1}{2}\theta^2$. Also

$$\frac{R}{R+h} = 1 - \frac{h}{R+h} \approx 1 - \frac{h}{R},$$

since $R + h \approx R$. Equating,

$$1 - \tfrac{1}{2}\theta^2 \approx 1 - \frac{h}{R} \quad\Longrightarrow\quad \theta^2 \approx \frac{2h}{R} \quad\Longrightarrow\quad \theta \approx \sqrt{\frac{2h}{R}},$$

taking the positive root as $\theta > 0$. Hence

$$d = R\theta \approx R\sqrt{\frac{2h}{R}} = \sqrt{2}\,(Rh)^{1/2},$$

which is the required form with $k = \sqrt{2}$.

Part (iii)

Now $R/(R+h)$ is small, so $\cos\theta$ is near 0 and θ is close to $\frac{\pi}{2}$, not small. The given approximations only work for small angles, so we measure the angle that *is* small.

Ask yourself Which angle is small here, and which approximation applies to it?

Write $\theta = \frac{\pi}{2} - \phi$, where $\phi = \angle OXZ$ is the angle at the satellite, which is small when the satellite is far away. Then $\cos\theta = \sin\phi$, so

$$\sin\phi = \frac{R}{R+h}.$$

We need this to second order in R/h, because the answer has an R^2/h term. By the binomial expansion, valid as $R/h < 1$,

$$\frac{R}{R+h} = \frac{R}{h}\left(1 + \frac{R}{h}\right)^{-1} = \frac{R}{h} - \frac{R^2}{h^2} + \cdots.$$

With $\sin\phi \approx \phi$,

$$\phi \approx \frac{R}{h} - \frac{R^2}{h^2}.$$

Therefore

$$\begin{aligned} d = R\theta &= R\left(\frac{\pi}{2} - \phi\right) \\ &\approx \frac{\pi}{2}R - \frac{R^2}{h} + \frac{R^3}{h^2}. \end{aligned}$$

The last term is smaller than R^2/h by a further factor of R/h, so we drop it, giving

$$d \approx \frac{\pi}{2}R - \frac{R^2}{h},$$

which is the required form with $a = \frac{\pi}{2}$ and $b = -1$.

Using $\sin\phi \approx \phi$ is safe to this order: the error in it is about $\frac{1}{6}\phi^3$, which is of order R^3/h^3, so it changes d only at order R^4/h^3.

Checks

- Limits make sense. As $h \to 0$, $d \to 0$. As $h \to \infty$, $d \to \frac{\pi}{2}R$, a quarter of the way round the Earth: a very distant satellite sees almost a full hemisphere. The term $-R^2/h$ is negative, so d approaches $\frac{\pi}{2}R$ from below, as it must since $\theta < \frac{\pi}{2}$.
- Numerically, with $R = 1$ and $h = 10^{-4}$, the exact d divided by $\sqrt{2Rh}$ is 0.99996. With $h = 1$ and $R = 0.01$, the exact d and $\frac{\pi}{2}R - R^2/h$ differ by 9.9×10^{-7}, which is about R^3/h^2, as predicted.
- For a satellite 300 km above an Earth of radius 6370 km, the exact d is about 1918 km and $\sqrt{2Rh}$ gives 1955 km.

Common slips

- Using $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ in part (iii), where θ is near $\frac{\pi}{2}$ and not small.
- Expanding $R/(R+h)$ only to first order in part (iii), which loses the R^2/h term and gives the wrong b.

Beyond STEP

Both parts are asymptotic expansions: the exact d is expanded in powers of a small parameter, h/R or R/h, and we keep terms up to a stated order while bounding the error. Making "$\approx$" precise with big-O notation and Taylor's theorem with remainder is a first-year analysis topic; see Abbott, *Understanding Analysis*. Siklos, *Advanced Problems in Mathematics*, has more STEP practice in this kind of approximation.

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